Sign-safe Kummer differential walk

Reference invariant gate · 2026-10-22 UTC

Scope. 21 Scholten models and 1,260 frozen public targets (60/model).

Observed. 1260 source scalars checked; 8820 target

coefficient-pair cases and 630 model exception cases passed.

Failures: 0 targets; 0 exceptions.

Compatible sign lifts: 1 in 1,260 cases; 2 in 7,560 cases.

Every compatible set produced one quotient output; the signed anchor

selected one Jacobian lift. Raw rows and a byte-identical rerun are retained.

Method. K(D) is canonical min(D,-D). Differential outputs were

enumerated from Jacobian lifts using K(A), K(B), K(A-B). A signed G

and K(A+B+G) selected the unique signed output.

Boundary. This uses Jacobian addition as the reference oracle. It does

not implement coordinate-only Kummer arithmetic or a DLP collector.

Candidate/rho ratios in all 36 price cells are null; 0 of 1,260

requested Kummer full-collector targets were executed.

A fixed-offset sidecar updated by one xADD per jump J needs S-J in S;

this restricted design requires at least order(J) stored offsets.

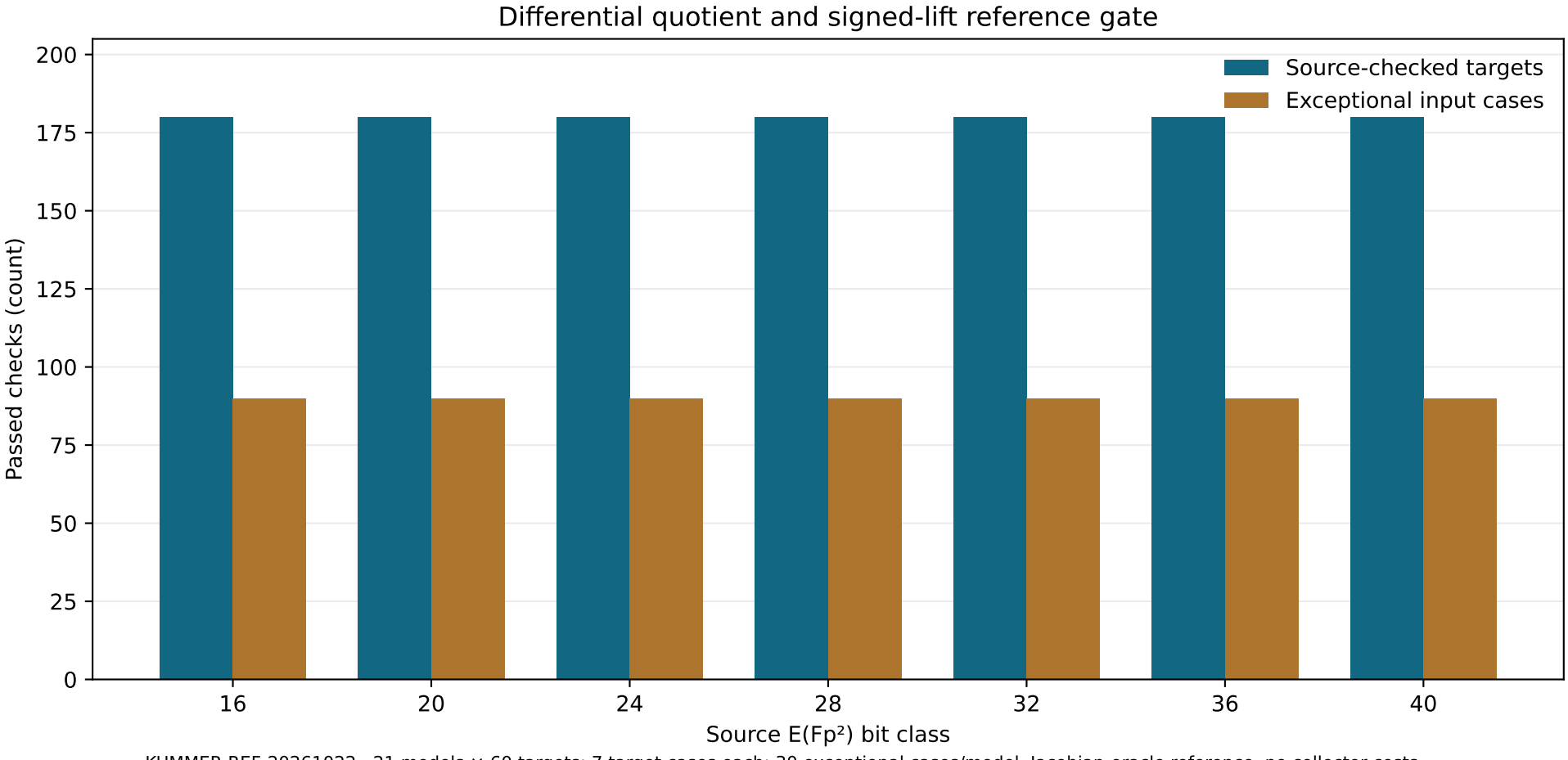
Source records: results/target-gate-*.jsonl, model-exceptions-*.jsonl,

summary.json, price-grid.csv, manifest.json and SHA256SUMS.

Primary arithmetic reference: Chung, Costello & Smith, ePrint 2015/983.

https://eprint.iacr.org/2015/983.pdf

Gaudry fast Kummer: https://eprint.iacr.org/2005/314.pdf


Differential quotient and signed-lift reference gate
Source-checked targets
Exceptional input cases
Passed checks (count)
0
25
50
75
100
125
150
175
200
16
20
24
28
32
36
40
Source E(Fp²) bit class
KUMMER-REF-20261022 · 21 models × 60 targets; 7 target cases each; 30 exceptional cases/model. Jacobian-oracle reference, no collector costs.